

HCSM-13

The Processor T and the Fermion/Boson-Class Correspondence

The Rank Ten Projection, the F/B Split,
and the Localization of Chirality Breaking
The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic eight by eight torus. The fourteen modes at the self paired eigenvalue of the discrete Laplacian are identified with one Standard Model generation. On this multiplet, the processor T is the rank ten orthogonal projection that separates the ten Standard Model fermions from the four Standard Model bosons. This paper proves two theorems about the processor and its action. First, the processor commutes with the class-C projector of the winding partition, and its image is exactly the union of winding classes A and B, while its kernel is exactly winding class C. Equivalently, $T = I - P_C$, so the fermion/boson split is the coarse-graining of the winding class partition that identifies classes A and B. Second, the processor T and the chiral reflection rs fail to commute only on the two dimensional plane spanned by the dark photon and the tau; on the other twelve modes, T and rs commute. On the two dimensional plane of the failure, the commutator is a $\mathbb{Z}_4$ rotation generator. This is the framework's chirality breaking: it is localized to a single two mode plane. All theorems are HCSM-native, and all numerical claims are verified at machine precision.

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1 Introduction

1.1 The processor in HCSM

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic eight by eight torus [1]. The fourteen modes at the self paired eigenvalue of the discrete Laplacian are identified with one Standard Model generation. Their dihedral representation theory determines the gauge algebra; their mass-ordering labels determine the mass spectrum; the zero state vector from the fourteen by nineteen measurement matrix determines the flavor structure.

Above the substrate, the processor T is the machine that reads the fourteen mode multiplet and separates it into a fermion sector and a boson sector. The processor is a rank-ten orthogonal projection: it keeps ten modes (the Standard Model fermions) and kills four modes (the Standard Model bosons). Its matrix is diagonal in the mode basis.

This paper proves that the processor's output is not an independent structure on the multiplet. It is the coarse-graining of the winding class partition that identifies two of the three winding classes. And the failure of the processor to commute with the framework's chiral reflection is localized to a single two-mode plane.

1.2 The one-line summary

The processor T is exactly the complement of the class-C projector. The fermion/boson split is the same as the winding class partition, with classes A and B fused into a single fermion sector. And chirality is broken only on the plane spanned by the dark photon and the tau: twelve of the fourteen modes see no chirality breaking at all.

1.3 The two theorems of this paper

The paper is organized around two theorems.

1. **Fermion/Boson-Class Correspondence Theorem** (Section 3). The processor T commutes with the class-C projector P_C . Its image is exactly the union of winding classes A and B, and its kernel is exactly winding class C. Equivalently, $T = I - P_C$. The fermion/boson split is the coarse-graining of the winding class partition that identifies classes A and B.
2. **Chirality-Breaking Localization Theorem** (Section 4). The commutator $[T, P_{rs}]$ of the processor with the chiral reflection P_{rs} is supported exactly on the two element set consisting of the dark photon and the tau. On the two-dimensional plane spanned by these two modes, the commutator is a $\mathbb{Z}_4$ rotation generator; on the other twelve modes, T and rs commute.

1.4 Cross-references to the HCSM paper stack

The paper depends on three companion papers in the HCSM stack, whose numbering follows the master index in HCSM-00:

- **HCSM-04** [2]: the fourteen-mode multiplet and its dihedral orbit decomposition.
- **HCSM-14** [3]: the winding generator spectrum and the winding class partition.
- **HCSM-26** [5]: the observer pair, which supplies the chiral reflection rs and the fixed-locus theorem.

Additional references to HCSM-22 (the zero state) and HCSM-27 (Klein orbit structure) appear where needed.

1.5 Organization

Section 2 states the setup: the fourteen-mode multiplet, the winding class partition, the processor T , and the chiral reflection rs . Section 3 proves the Fermion/Boson Class Correspondence Theorem. Section 4 proves the Chirality-Breaking Localization Theorem. Section 5 discusses the structural interpretation. Section 6 gives reproducibility notes. Section 7 concludes.

2 Setup

This section states the HCSM primitives on which the paper depends. All statements are HCSM-native; the reader is referred to HCSM-00 for the full foundations.

2.1 The fourteen-mode multiplet

The substrate is the periodic eight-by-eight torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites [1]. The discrete five-point Laplacian has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (1)$$

The self-paired eigenvalue $\lambda = 4$ has multiplicity fourteen. In the standard mode order used throughout HCSM (matching the zero-state vector of HCSM-22 [4]), the fourteen modes are

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t). \quad (2)$$

2.2 The winding class partition

The winding generator H_{wind} has spectrum $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$. The fourteen modes partition into three classes by frequency [3]:

$$A = \{\nu, e, t\}, \quad \text{size } 3, \quad (3)$$

$$B = \{u, d, s, \mu, c, \tau, b\}, \quad \text{size } 7, \quad (4)$$

$$C = \{\text{dark}, W, Z, H\}, \quad \text{size } 4. \quad (5)$$

The class projectors P_A, P_B, P_C are diagonal in the mode basis, mutually orthogonal, idempotent, and sum to the identity.

2.3 The processor T

Definition 2.1 (Processor). The *processor* T is the rank-ten orthogonal projection onto the fermion sector F :

$$T = P_{\text{mesh}} + P_{\tau}, \quad (6)$$

where P_{mesh} projects onto the nine mesh resonances (those modes with $\delta \neq 0$) and P_{τ} projects onto the unique mode with zero-state amplitude $s = +1$.

In the standard mode order, the processor is diagonal with entries:

$$\text{diag}(T) = (1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 1). \quad (7)$$

The fermion sector and boson sector are

$$F = \{\nu, e, u, d, s, \mu, c, \tau, b, t\}, \quad |F| = 10, \quad (8)$$

$$B = \{\text{dark}, W, Z, H\}, \quad |B| = 4. \quad (9)$$

Proposition 2.2 (Processor properties). *The processor satisfies $T^2 = T$, $T^\dagger = T$, and $\text{rank}(T) = 10$.*

Proof. Orthogonal projection by construction. Verified at machine precision in the companion verification script. $\square$

2.4 The chiral reflection rs

The chiral reflection is the map $rs : (n_1, n_2) \mapsto (n_2, n_1) \pmod{8}$. It is an element of the dihedral point group of the square, and it satisfies $rs^2 = e$. It has been identified in HCSM-26 as the unique chiral reflection of the dihedral group: the unique element that fixes exactly the pair $\{(2, 2), (6, 6)\}$ on the fourteen-mode multiplet and anticommutes with the winding generator.

Let P_{rs} denote the permutation matrix acting on $\mathbb{C}^{14}$ that implements rs in the mode basis. It satisfies $P_{rs}^2 = I$, and it is Hermitian.

3 The Fermion/Boson-Class Correspondence

This section proves that the processor T is exactly the complement of the class-C projector.

3.1 The complement formulation

Theorem 3.1 (Fermion/Boson-Class Correspondence). *The processor T commutes with the class-C projector P_C and equals*

$$T = I - P_C = P_A + P_B. \quad (10)$$

Equivalently:

1. The image of T is exactly the union of winding classes A and B : $\text{image}(T) = A \cup B$.
2. The kernel of T is exactly winding class C : $\ker(T) = C$.
3. $[T, P_C] = 0$.

Proof. Direct inspection of the diagonal entries of T and of the class membership of each mode.

The processor T is diagonal, with $\text{diag}(T) = (1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 1)$ in the standard mode order. Hence:

- The image of T (modes with diagonal entry 1) is

$$\text{image}(T) = \{\nu, e, u, d, s, \mu, c, \tau, b, t\} = F. \quad (11)$$

- The kernel of T (modes with diagonal entry 0) is

$$\ker(T) = \{\text{dark}, W, Z, H\} = B. \quad (12)$$

On the other hand, the winding classes are

$$A = \{\nu, e, t\}, \quad (13)$$

$$B = \{u, d, s, \mu, c, \tau, b\}, \quad (14)$$

$$C = \{\text{dark}, W, Z, H\}. \quad (15)$$

Hence $A \cup B = \{\nu, e, u, d, s, \mu, c, \tau, b, t\} = F = \text{image}(T)$ and $C = \{\text{dark}, W, Z, H\} = B = \ker(T)$.

Since T is diagonal and P_C is diagonal, they commute. The identity $T = I - P_C$ is verified by comparing $\text{diag}(I - P_C) = (1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 1)$ with $\text{diag}(T)$.

The full matrix identity $T = I - P_C$ is verified at machine precision:

$$\|T - (I - P_C)\|_F = 0.000 \times 10^0. \quad (16)$$

□

Corollary 3.2 (The fermion/boson split is a coarse-graining). *The fermion/boson split is the coarse-graining of the winding class partition that identifies classes A and B . Equivalently, the fermion sector is the image of the processor, and the boson sector is the kernel of the processor.*

Proof. Immediate from the theorem. □

Remark 3.3. The theorem states that the processor does not add structure beyond the winding class partition. The winding class partition is finer grained: it separates class A from class B . The fermion/boson split ignores that distinction. The winding class partition and the fermion/boson split encode the same physical content, viewed at different coarse graining scales.

3.2 The Landauer channel

Proposition 3.4 (Landauer entropy reduction). *The processor reduces the entropy of the maximally mixed state on the fifty six dimensional Hodge sector to the entropy of the maximally mixed state on the fermion/boson product. The entropy reduction is*

$$\Delta S = \log 56 - (\log 10 + \log 4) = \log 56 - \log 40 = \log(7/5) = 0.33647 \dots \text{ nats}. \quad (17)$$

This saturates the deterministic Landauer bound.

Proof. Direct computation. □

4 The Chirality Breaking Localization Theorem

This section proves that the processor T and the chiral reflection rs fail to commute only on the two-dimensional plane spanned by the dark photon and the tau.

4.1 The commutator

Theorem 4.1 (Chirality Breaking Localization). *The commutator of the processor with the chiral reflection,*

$$C_{rs} = T \cdot P_{rs} - P_{rs} \cdot T, \quad (18)$$

is supported exactly on the two-element set

$$\{\text{dark}, \tau\}. \quad (19)$$

On every other pair of indices, the commutator vanishes. The two dimensional plane spanned by the dark photon and the tau is the framework's unique chirality-breaking plane.

Proof. Direct computation of the commutator.

The processor T is diagonal with $\text{diag}(T) = (1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0, 1)$. The chiral reflection P_{rs} permutes the modes by the diagonal reflection. We compute $(T \cdot P_{rs})_{ij} = T_{ii} \cdot (P_{rs})_{ij}$ and $(P_{rs} \cdot T)_{ij} = (P_{rs})_{ij} \cdot T_{jj}$.

For the commutator to be nonzero at entry (i, j) , we need both $(P_{rs})_{ij} = 1$ (so rs maps mode j to mode i) and $T_{ii} \neq T_{jj}$.

The chiral reflection maps each mode as follows in the standard mode order:

$$\begin{array}{ll} rs(\nu) = e, & rs(e) = \nu, \\ rs(u) = s, & rs(s) = u, \\ rs(d) = b, & rs(b) = d, \\ rs(\mu) = c, & rs(c) = \mu, \\ rs(\tau) = \text{dark}, & rs(\text{dark}) = \tau, \\ rs(W) = W \text{ (fixed)}, & rs(Z) = H, \\ rs(H) = Z, & rs(t) = t \text{ (fixed)}. \end{array}$$

For the pairs where rs swaps two modes, we examine whether their diagonal entries in T differ:

$$\begin{array}{ll} \nu, e : T_{\nu\nu} = 1, T_{ee} = 1 & \text{(equal)}, \\ u, s : T_{uu} = 1, T_{ss} = 1 & \text{(equal)}, \\ d, b : T_{dd} = 1, T_{bb} = 1 & \text{(equal)}, \\ \mu, c : T_{\mu\mu} = 1, T_{cc} = 1 & \text{(equal)}, \\ \tau, \text{dark} : T_{\tau\tau} = 1, T_{\text{dark}, \text{dark}} = 0 & \text{(different)}, \\ Z, H : T_{ZZ} = 0, T_{HH} = 0 & \text{(equal)}. \end{array}$$

The only pair with a nonzero commutator is $\{\text{dark}, \tau\}$. The two fixed points W and t contribute nothing, since rs acts trivially on them and hence cannot create off-diagonal commutator entries.

Hence the commutator is supported exactly on the pair $\{\text{dark}, \tau\}$. $\square$

4.2 The $\mathbb{Z}_4$ structure of the commutator

Proposition 4.2 (The commutator as a $\mathbb{Z}_4$ generator). *On the two-dimensional plane spanned by the dark photon and the tau, the commutator C_{rs} is*

$$C_{rs}|_{\{\text{dark}, \tau\}} = \begin{pmatrix} 0 & -1 \\ +1 & 0 \end{pmatrix}. \quad (20)$$

Its square is $-P_{\{\text{dark}, \tau\}}$, and its fourth power is $+P_{\{\text{dark}, \tau\}}$. Hence C_{rs} generates a $\mathbb{Z}_4$ rotation on the chirality-breaking plane.

Proof. Direct computation. The $(1, 1)$ entry of the commutator on the plane is $T_{\tau\tau} - T_{\text{dark}, \text{dark}} = 1 - 0 = 1$. The $(2, 2)$ entry is $T_{\text{dark}, \text{dark}} - T_{\tau\tau} = 0 - 1 = -1$. Both vanish. The off-diagonal entries are $-(P_{rs})_{\text{dark}, \tau} \cdot T_{\tau\tau} = -1$ and $+(P_{rs})_{\tau, \text{dark}} \cdot T_{\text{dark}, \text{dark}} = +1 \cdot (P_{rs})_{\tau, \text{dark}} \cdot (1 - 0) = +1$. Hence

$$C_{rs}|_{\{\text{dark}, \tau\}} = \begin{pmatrix} 0 & -1 \\ +1 & 0 \end{pmatrix}. \quad (21)$$

Squaring:

$$C_{rs}^2|_{\{\text{dark}, \tau\}} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -P_{\{\text{dark}, \tau\}}. \quad (22)$$

Fourth power:

$$C_{rs}^4|_{\{\text{dark}, \tau\}} = \begin{pmatrix} +1 & 0 \\ 0 & +1 \end{pmatrix} = +P_{\{\text{dark}, \tau\}}. \quad (23)$$

The commutator therefore generates a $\mathbb{Z}_4$ rotation on the chirality breaking plane, verified at machine precision in the companion verification script. $\square$

4.3 Commutation on the other twelve modes

Corollary 4.3 (Commutation elsewhere). *The processor T and the chiral reflection rs commute on the twelve modes not in the chirality-breaking plane $\{\text{dark}, \tau\}$.*

Proof. Immediate from Theorem 4.1. The restriction of the commutator to the subspace spanned by the other twelve modes is identically zero. $\square$

Remark 4.4. The localization of chirality-breaking is sharp. On the twelve modes outside the plane $\{\text{dark}, \tau\}$, the processor preserves the chiral reflection's action exactly. Only on the two-mode plane does chirality break, and the breaking is a $\mathbb{Z}_4$ rotation. This makes the framework's chirality content highly constrained.

5 Discussion

5.1 What the processor is

The processor T is not an independent structure on the fourteen-mode multiplet. It is the complement of the class-C projector. Its image is the union of classes A and B; its kernel is class C. The fermion/boson split is exactly the coarse graining of the winding class partition that fuses classes A and B.

This means that the processor's output is determined by the winding class partition, which in turn is determined by the dihedral orbit partition plus the fermion/boson homogeneity constraint of HCSM-28. The whole processor structure is therefore determined by the substrate, without independent input.

5.2 What the chirality breaking is

The chirality-breaking of the processor is not a global feature. It is localized to the plane $\{\text{dark}, \tau\}$. On the twelve other modes, the processor and the chiral reflection commute exactly.

The two chirality breaking modes are the dark photon (a boson in the kernel of T) and the tau (a fermion in the image of T). Together they form the unique two-dimensional subspace on which the processor breaks the symmetry.

This two mode plane is the intersection of the processor with the chirality structure: it is the unique place where the processor's fermion/boson distinction conflicts with the dihedral reflection.

5.3 Relation to the observer pair

The observer pair of HCSM-26 is the pair of fixed points of the chiral reflection: the W boson and the top quark. These are precisely the two modes on which the chiral reflection acts trivially. Since T and rs commute on W and t , the observer pair is not part of the chirality-breaking plane.

The framework's structure therefore separates cleanly: the observer pair is the fixed locus of the chiral reflection, and the chirality breaking plane is the unique two mode plane where the processor and the chiral reflection disagree. These are two independent structural features of the fourteen-mode multiplet.

5.4 Relation to the Klein orbit structure

The two chirality breaking modes, the dark photon and the tau, are members of the four cycle orbit of the Klein group (HCSM-27). Specifically, the four cycle orbit is $\{\text{dark}, s, u, \tau\}$, and the chirality breaking pair is the subset $\{\text{dark}, \tau\}$. The chiral reflection and the half-period translation both individually swap these two modes with other modes in the four cycle; the product fixes them together, which is why the four-cycle orbit has trivial pointwise stabilizer.

6 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at the verification folder with the number thirteen. The script constructs the fourteen mode multiplet, the winding class projectors, the processor T , and the chiral reflection P_{rs} from HCSM-native definitions, and verifies:

1. The processor properties $T^2 = T$, $T^\dagger = T$, $\text{rank}(T) = 10$.
2. The matrix identity $T = I - P_C$ to machine precision.
3. The image $\text{image}(T) = A \cup B$ and the kernel $\ker(T) = C$.
4. The commutation $[T, P_C] = 0$.
5. The commutator $[T, P_{rs}]$ restricted to the pair $\{\text{dark}, \tau\}$ and zero on the other twelve modes.
6. The $\mathbb{Z}_4$ structure: $[T, P_{rs}]^2 = -P_{\{\text{dark}, \tau\}}$ and $[T, P_{rs}]^4 = +P_{\{\text{dark}, \tau\}}$.
7. The Landauer entropy reduction $\Delta S = \log(7/5)$.

Running the master verification script reproduces every numerical claim in the HCSM paper stack.

7 Conclusion

This paper has proven two theorems about the processor T of the Hodge Complex Standard Model.

The first is the Fermion/Boson-Class Correspondence Theorem: the processor is exactly the complement of the class-C projector of the winding partition. Its image is the union of winding classes A and B; its kernel is class C. The fermion/boson split is the coarse-graining of the winding class partition that fuses classes A and B. Equivalently, $T = I - P_C$.

The second is the Chirality-Breaking Localization Theorem: the processor's commutator with the chiral reflection is supported exactly on the two-mode plane spanned by the dark photon and the tau. On this plane, the commutator is a $\mathbb{Z}_4$ rotation generator. On the other twelve modes, the processor and the chiral reflection commute exactly.

The framework's chirality content is therefore highly constrained. The chirality breaking is localized to a single two-mode plane, and the observer pair of HCSM-26 (the W boson and the top quark) is exactly the complement: the fixed locus of the chiral reflection. The processor's output and its chirality breaking are both determined by the winding class partition and the dihedral reflection, with no independent input.

References

- [1] S. Preschutti, *HCSM-00: Foundations, Notation, and Mapping*, HCSM White Paper Series (2026).
- [2] S. Preschutti, *HCSM-04: The 14-Mode Multiplet at $\lambda = 4$* , HCSM White Paper Series (2026).
- [3] S. Preschutti, *HCSM-14: Winding Generator Spectrum and Class Partition*, HCSM White Paper Series (2026).
- [4] S. Preschutti, *HCSM-22: The Zero State*, HCSM White Paper Series (2026).
- [5] S. Preschutti, *HCSM-26: The Observer Pair*, HCSM White Paper Series (2026).
- [6] S. Preschutti, *HCSM-27: Klein Orbit Structure*, HCSM White Paper Series (2026).
- [7] S. Preschutti, *HCSM-28: Orbit-Class Reconstruction*, HCSM White Paper Series (2026).

A Web Crawler Metadata

A.1 Dublin Core

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A.2 Schema.org JSON-LD

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A.3 BibTeX for Automated Indexing

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    Theorem shows that the commutator of  $T$  with the chiral reflection  $rs$ 
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